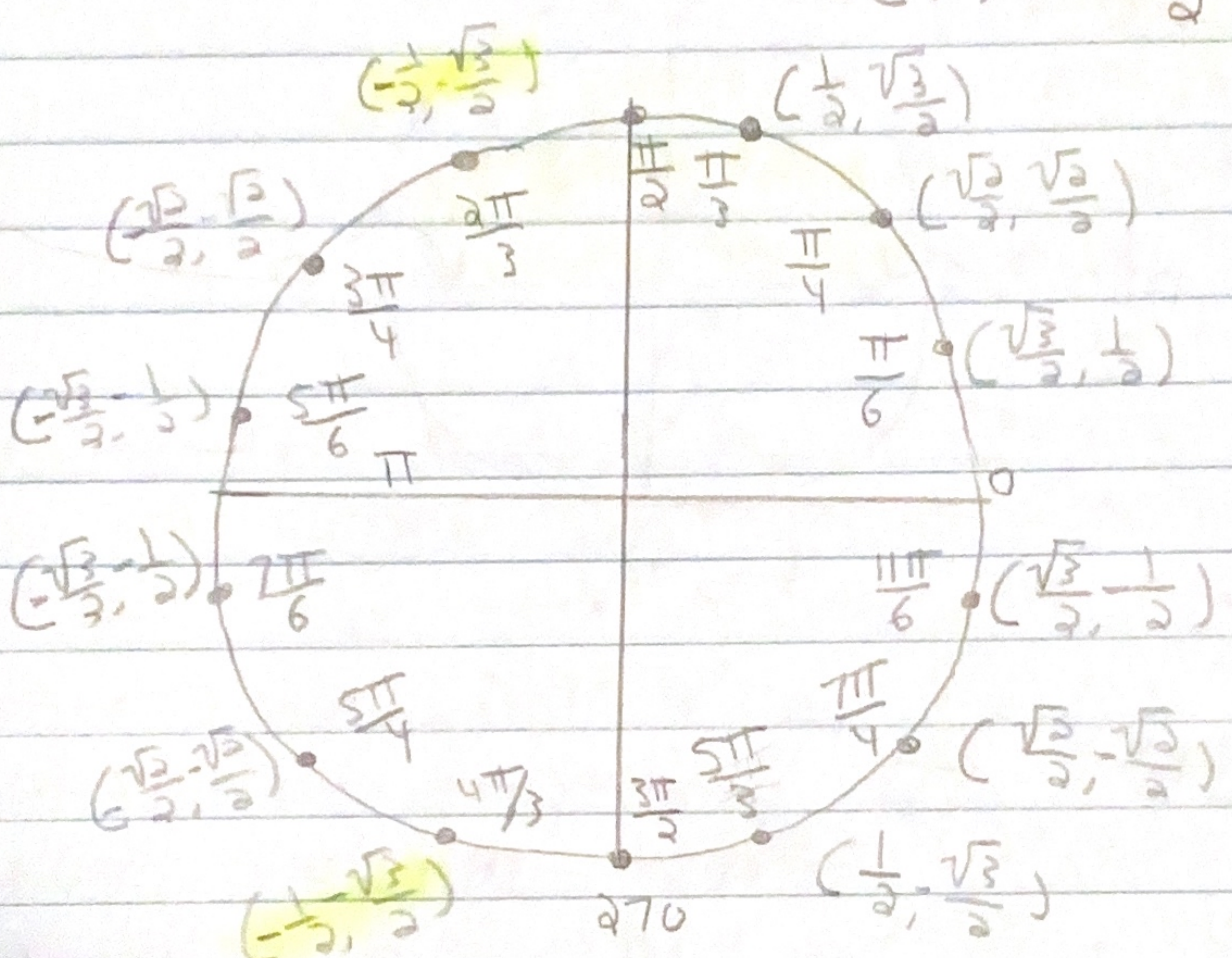
aug-23 trig function notes

find $\cos^{-1}(-\frac{1}{2})$ ← This is asking for you to find the radian that gets plugged into "cos" and gives you "$\frac{2\pi}{3}$" as a result

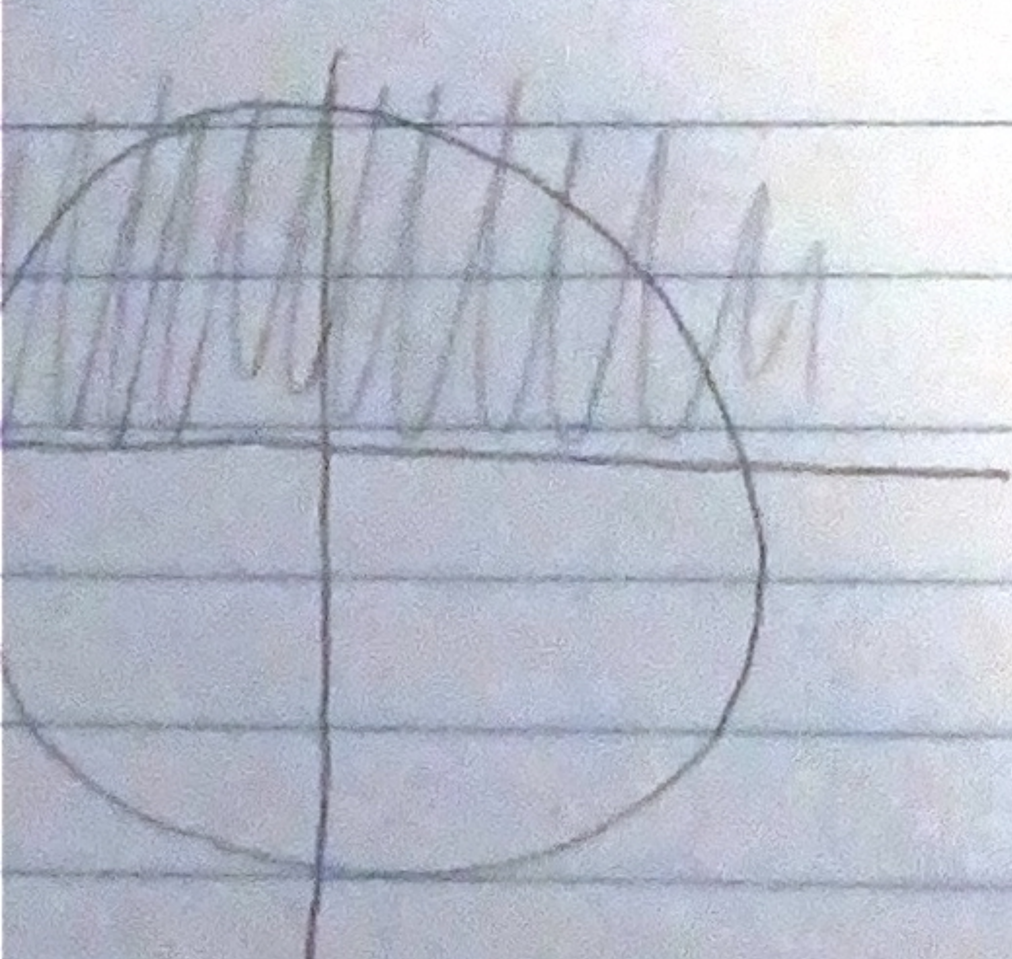
(cos, sin)

$\cos(?) = -\frac{1}{2}$



but there are two radian's that give you "$-\frac{1}{2}$" and that cannot work it can only work if it was a equation but the inverses is not an equation it must have one answer

So restriction were introducted



cos was restricted to the first two quadrent or $[0, \pi)$ if the answer is outside the restriction it cannot work

So $\cos^{-1}(-\frac{1}{2}) = \frac{2\pi}{3}$

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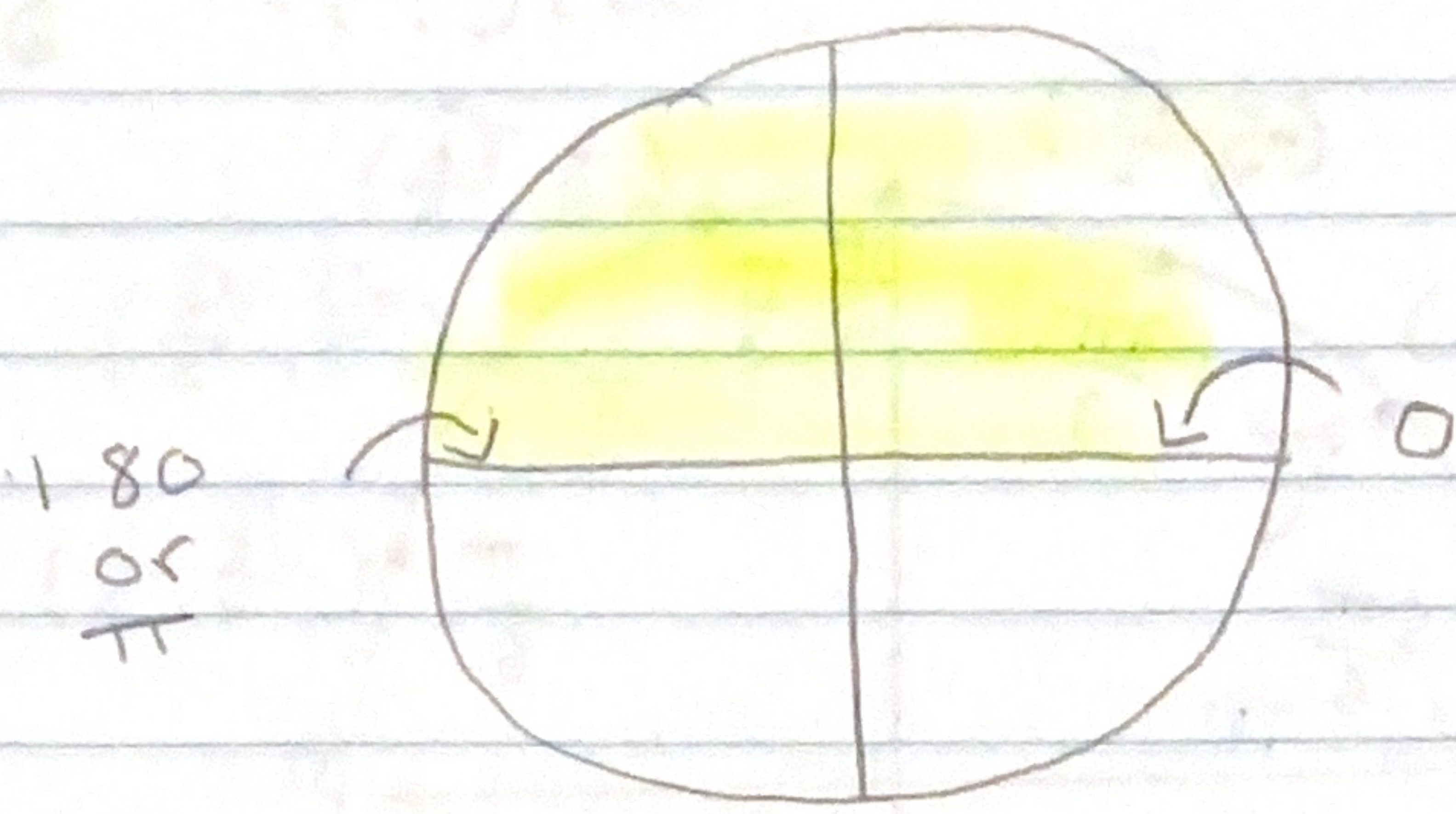
← So if you get an answer out of the restriction it cannot work

we can test this by $\cos(\frac{2\pi}{3}) = -\frac{1}{2}$

So the cos restriction are from $[0, \pi)$



$[0, \pi)$

- $\tan^{-1}(-1)$ ← you can think of this as $\tan\theta = -1$ but there is only one answer

- tan inverse has this as a restriction

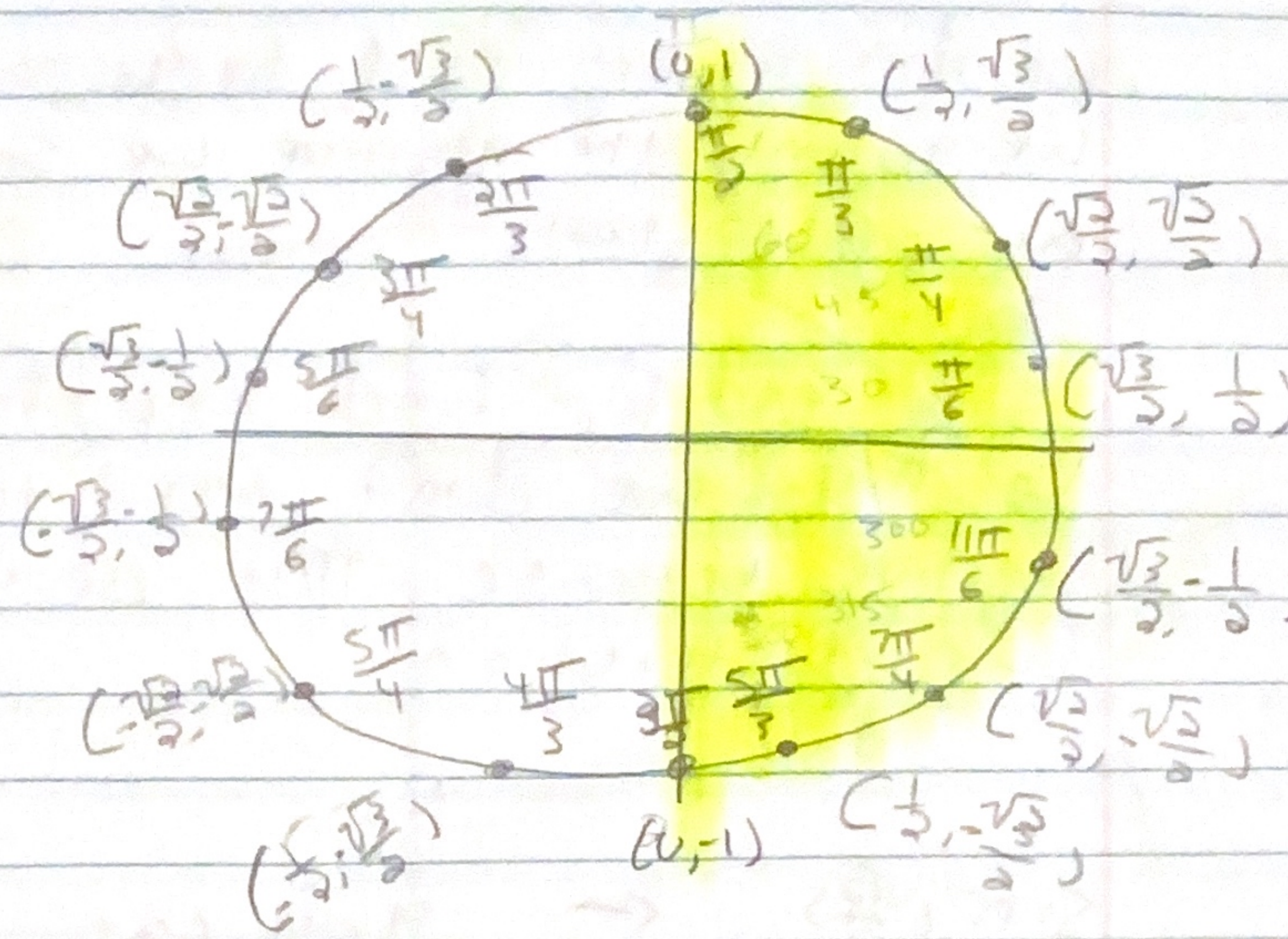
$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ or } (-90, 90)$$

So tanget cannot be $-\frac{\pi}{2}$ or $\frac{\pi}{2}$

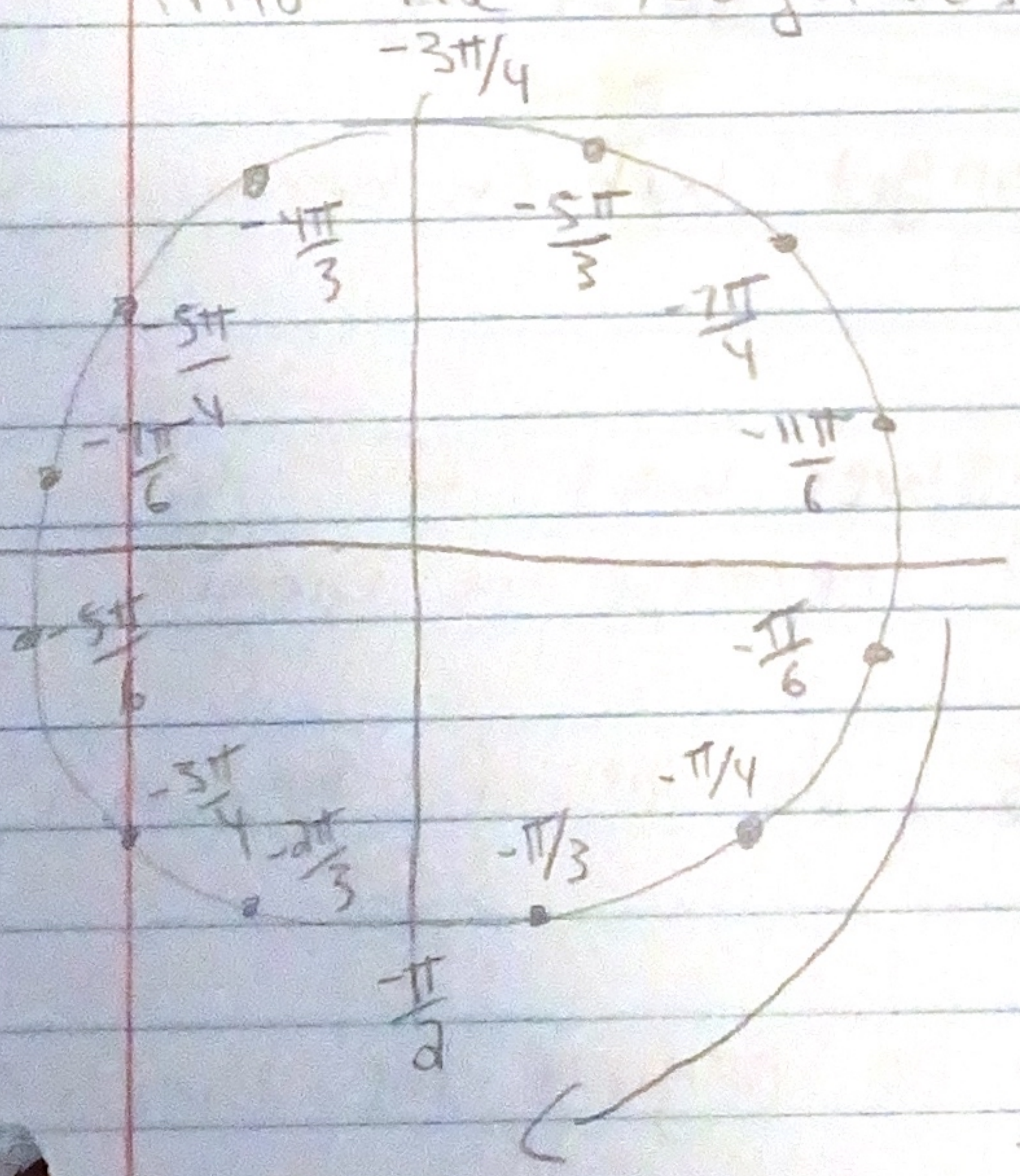
- this is what happen when tanget is $-\frac{\pi}{2}$ or $\frac{\pi}{2}$

$\tan = \frac{\sin}{\cos}$ $\quad \frac{\pi}{2} = (0, -1) \quad \frac{-1}{0}$ = cannot work

$\tan^{-1}(-1)$



The tanget restriction is highlighted and we can tell within that restriction our answer is "$\frac{7\pi}{4}$" but that is not correct that violate our restriction $\frac{7\pi}{4}$ or 315 is Not in our restriction so what can we do to get the answer well we can got into the Negitives



So our answer will be $-\pi/4$ we can check by doing this

$\tan\left(-\frac{\pi}{4} + \frac{7\pi}{4}\right) = \frac{8\pi}{4} = 2\pi$

add by a number that will get it equal to 2π

$\tan\left(\frac{7\pi}{4}\right) = -1$

the number that we add will give us $\frac{7\pi}{4}$ the same number that violated the restriction we don alway need to go into the negitive we go into the negitive we something violates our restriction

if we had to find the $\tan^{-1}(1)$ it would be $\pi/4$ we don't need to go into the negitive because $\pi/4$ or 45 is within our restriction

$\sin^{-1}(-\frac{\sqrt{2}}{2})$ ← you can think of this as $\sin\theta = -1$ just you only can have one answer

the restriction for sin is $[-\frac{\pi}{2}, \frac{\pi}{2}]$ or $[-90, 90]$ the same as the tanget exept sin can be either $-\frac{\pi}{2}$ or $\frac{\pi}{2}$

$\sin(\frac{\pi}{2}) = 1$ unlike tanget which become undef

if you thought the answer was "$\frac{7\pi}{4}$" you would be incorrect why? because it violate our restriction

$\frac{7\pi}{4}$ or 315 do not fit between our restrictions

So we must go into the negitive the answer will be $-\pi/4$

$\sin(-\pi/4 + \frac{7\pi}{4}) = \frac{8\pi}{2} = 2\pi$ $\sin(\frac{7\pi}{4}) = -\frac{\sqrt{2}}{2}$

negitive unit circle